I –Suggestions of Measure of Central Tendency:

- What is central tendency? Write down the main measure of central tendency?

- What are the measures of Central tendency? A frequency distribution of age groups of 102 persons are given below :

Age Group	Frequency
10-15	13
15-20	18
20-25	35
25-30	17
30-35	11
35-40	8

Calculate arithmetic mean, median and Harmonic mean. (RU-ICE: 2014)

- Define arithmetic mean, median and mode for the grouped data with their merit. (RU-ICE: 2015)
- Show that, for AM ≥ GM ≥ HM for 2 non-zero positive observations. (RU-EEE: 2014,16.19)

- For two positive observation show that $AH= G^2$; (RU-ICE : 2011)
Where, **A**= Arithmetic mean, **H**= Harmonic mean and **G**= Geometric mean

- Show that sum of squares of the deviation of a set of observation is minimum when it is taken from arithmetic mean. (RU-ICE: 2015).

- State various types of averages and write merits and demerits of each measure of central tendency. (RUET: 2011,2013).

- Show that, $\Sigma\delta_i (x_i-a)^2$ has a minimum value. (RUET: 2013.2016)
- Define geometric mean and harmonic mean with example .For the following frequency distribution, calculate mode and median ---

Class interval	Frequency
13-15	3
15-17	7
17-19	12
19-21	6
21-23	2

(RU-ICE: 2015).

- The frequency distribution given below the marks obtained in a exam by 80 Students are……..

Class interval	Frequency
50-60	5
60-70	9
70-80	13
80-90	20
90-100	19
100-110	9
110 and Above	5
	N=80

(RUET: 2011,15 -civil).

Compute, (a) A.M G.M and H.M
(b) Median, Q_1 , Q_3, D_4 and P_{80}
(C) Mode

- From the following data, compute the values of upper and lower quartiles, median, D6, P20. And Mode

Marks	No. of. Students
Below 10	5
10-20	25
20-30	40
30-40	70
40-50	90
50-60	40
60-70	20
Above 70	10
	N=260